



SES-DCB-5120-1 Deep Cycle Type

APPLICATION

The specification is applicable to basic performance, technical requirement testing method, warning and caution of the Lithium iron phosphate rechargeable battery

GENERAL SPECIFICATION

16IFR32700-D12D6H4-48V100AhOuter Shell

Dimension(mm)

19inch chassis: 483(L)×386(W)×223 (H) ±0.5mm



PANEL INTERFACE

Name	Description	Note
Handle	Easy to move and install	
Wiring Terminal	Used for battery external output/input Used for the system module installed and fixed it is recommended to install on a 19-inch rack	Two sets of parallel outputs
Switch	Address allocation when system expands the application	
Run	LED tums to green when normal running(Green light)	
RS485	System communication RS485 interface	
Reset	Emergency system reset,to ensure maintainability	
Alarm	Alarm indication when system is abnormal	
Capacity Indication	Battery capacity estimation	4pcs green lights means 4 step capacity
PUSE Fuse	Overcurrent protection,the load current is too large exceeding the fuse range, the fuse is blown	

MECHANICAL CHARACTERISTICS

Items	Test Method and Condition	Test Tools	Criteria
Vibration Test	After standard charge, Vibrate cell in 1.6mm amplitude and frequency varied at 1 Hz/min. between10 to 55 Hz and return in within 30 minutes per axis of X,Y,Z axes	Secondary batteries test equipment,Vibration platform	No Fire No Smoke No Explosion
Short-circuit test	Keep short circuit for 1 hour with 100mΩ resistance after standard charge	Digital voltage meter, Secondary Batteries Testing equipment	No Fire No Explosion
High temperature and high humidity test	Keep battery under condition as 40°C±2°C Ambient Temperature & Humidity as 90%-95%	Thermometer Hygrometer	Discharge time ≥ 36min No distortion No explosion

BATTERY PERFORMANCE

Nominal Capacity	100Ah	
Nominal Voltage	51.2V	
Nominal Energy	5120Wh	
Max Charge Voltage	56.8V	CC to CV
Terminal Discharge Voltage	44.0V	
Standard Charge Current	0.1C	
Standard Discharge Current	0.33C	
Max Continuous Charge Current	0.5C	
Max Continuous Discharge Current	0.5C	
Cycle Life	>1600 times	80%DoD@0.33C
Weight	52Kg	
Working Temperature	Charge,0°C~45°C	
	Discharge,-10°C~55°C	
Storage Temperature	-20~45°C	≤one month
	-20~35°C	≤three month
	-20~25°C	≤one year
	0~25°C	>one year

CELL SPECIFICATIONS

Items	Specifications	Test tools	Comments
Max Charge Voltage	3.55±0.05V	Voltage Meter	
Min Discharge Voltage	2.5±0.05V	Voltage Meter	
Initial Capacity	Nominal Capacity 6.0Ah@0.2C5A	Secondary batteries testing equipment	
AC Impedance	≤12mΩ	Impedance test equipment	AC Impedance 1KHz
Charge Standard	0.2C5A CC charge to Max charge voltage then CV charge till charging current decline to ≤0.01C	Digital voltage meter secondary batteries testing equipment	CC=Constant Current CV=Constant Voltage (3.60V)
Charge Time	Standard: 7.5hours (Ref)	Secondary batteries testing equipment	
Discharge Standard	0.2C5A CC discharge to discharge ending voltage	Digital voltage meter secondary batteries test equipment	
Cell Dimension	Height:71.1mmMax Diameter:32.3mmMax	Digital Calipers	Accurate sizes to ±0.25mm



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PCM PAEAMETER

Item	Description	Refer Standard
Over-charge protection	single cell over-charge alarm voltage	3.50~3.90V
	Total over-charge alarm voltage	56.40~57.60V
	Single cell over-charge test voltage	3.50~3.90V
	Total over-charge test voltage	56.40~57.60V
Over-discharge protection	Single cell over-discharge alarm voltage	2.00~2.90V
	Total over-discharge alarm voltage	36.00~48.00V
	Over-discharge test voltage	2.00~2.90V
	Total over-discharge test voltage	36.00~48.00V
Short circuit protection	Protect condition	External circuit short circuit
	Test delay time	<1ms
	Resume condition	Load disconnect If the short circuit is triggered multiple times, the lock is in the protection state.The user needs to check the external circuit, reset the BMS to release the protection
Temperature protection	High-temp.Alarm condition	35°C~65°C
	High-temp. Protect condition	35°C~65°C
	Low-temp.Alarm condition	-20°C~0°C
	Low-temp. Protect condition	-20°C~0°C
Charge balance	Balanced opening voltage	3.40V+0.02V
	Balanced opening pressure difference	50MV
	Balanced Current	85+20mA
Communication	RS485	Default baud rate :9600bit/S
Consumption	Working consumption current	≤50mA
	sleeping consumption current	≤300μA

BATTERY PERFORMANCES TESTING

Items	Test Method Condition	Test Tools	Criteria
Rated Capacity	a. (standard charge)charge the battery with standard CC to max charge voltage,change to CV charge.When the current less than 0.01C5A,stop charge,shelve30min b. (standard discharge)discharge the battery with standard current to terminal discharge voltage c. Check battery capacity	Secondary batteries testing equipment	Capacity≥95%
Cycle Life	a. Charge the battery with standard CC to max charge voltage, change to CV charge.When the current less than 0.01C5A, stop charge,shelve30min b. discharge the battery with standard current to terminal discharge voltage c. Record discharge capacity	Secondary batteries testing equipment	Rated cycles≥2000times
Self-discharge	a. After standard charge,stored at 25±5°C for 28 days, then discharge by standard current b. Check and record discharge capacity	Secondary batteries testing equipment	Capacity≥90% Self-discharge ≤10%
Temperature Characteristic	a. at 25±5°C, after standard charge,check and record capacity under the below temperature: 25±5°C, -10°C 60°C b. Count the percentage of discharge capacity at 25°C	Secondary batteries testing equipment Thermostat	25±°C,100% -10°C,≥60% 60°C,≥85%



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PACKAGING

The sketch, sizes, color of marking should match GB/T191-2000 requests.

- Model and specification of product;
- Quantity;
- Measure up marking;
- Manufacturing date
- Other markings (color.etc).

APPEARANCE

Scratch, flaw, crack, and leakage are not allowed

CAUTIONS IN USE

Please read the manual carefully before using it to ensure properly use.

- Do not make the battery exposure or thrown into fire
- Never reverse charge the battery.
- Never short circuit the battery.
- Avoid excessive physical shock or vibration.
- Do not disassemble or deform the battery.
- Never allow the battery to get wet or be immersed in water
- Do not use different types together.
- Keep away from children.
- Charge at the appropriate conditions.
- Never use the faulty charger to charging.
- Never keep charging more than 24 hours

STANDARD TEST CONDITIONS

Unless otherwise specified, all tests should be conducted within one month of delivery under the following conditions Temperature : $20\pm 5^{\circ}\text{C}$
Humidity: $60\pm 15\%\text{RH}$ Barometric: 86kpa-106kpa

STORAGE

- Store the battery in cool, dry and well-ventilated conditions.
- Store the battery in a individual room, separate from the other carriage.
- Regulations vary for different countries. Dispose of in accordance with local regulations.
- Please store the battery in the adequate temperature as mentioned in specifications and recharge if keep in storage more than 6 months.

BATTERY OPERATING INSTRUCTION

Charge

- Charge current: Never out of the max charge current as mentioned in specification.
- Charge voltage: Never out of the max charge voltage as mentioned in specification.
- Charge temperature: Please refer to the temperature range as specification.
- Charge as constant current before constant voltage never reverse charge the battery.

Discharge Current

The discharge current is not allowed to out of max current as specification. Otherwise, the battery will be over heat and capacity fading.

Discharge Temperaturet

Please refer to the temperature range as specification.

Over-Discharge

It's workable if over charge and discharge for a short while but not allow to do it for a long time. over discharge may result in self-energy disappear. Please keep a certain electric quantity to prevent over discharge.

WARRANTY PERIOD

Quality assurance for 2 year against manufacturing defects, but we are not responsible for the damage caused by inadequate and improper use. The information (subject to change without prior notice) contained in this document is for reference only and should not be used as a basis for product guarantee or warranty. For applications other than those described here, please contact our office. Manufacturer reserves the right to alter, amend the design, model and specification without prior notice.

OTHER CHEMICAL REACTION

The battery performance will reduce if over time using or unused for a long time due to it's a reaction of chemical. In addition, the battery life will be shorten or injury or damage itself from electrolyte leakage heating ignition or explosion for improper handling. It's necessary to replace battery if unable to charge even with proper way.

NOTE

Any other items which are not covered in this specification shall be agreed by both parties.